

Problem 1 : Hölder Continuous Functions

Let $f : [0, 1] \rightarrow \mathbb{R}$. We say that f is Hölder continuous of order $\alpha \in (0, 1)$ and write $f \in C^\alpha([0, 1])$ if

$$\|f\|_{C^\alpha} := \sup\{|f(x)|; x \in [0, 1]\} + \sup\left\{\frac{|f(x) - f(y)|}{|x - y|^\alpha}; x, y \in [0, 1], x \neq y\right\} < \infty.$$

For $f, g \in C^\alpha([0, 1])$, define $d(f, g) = \|f - g\|_{C^\alpha}$.

(a) Show that $(C^\alpha([0, 1]), d)$ is a complete metric space.

(b) Let (f_n) be a sequence in $C^{1/2}([0, 1])$ such that $\|f_n\|_{C^{1/2}} \leq 1$ for $n = 1, 2, \dots$. Show that (f_n) has a subsequence that converges in $C^{1/3}([0, 1])$.

Solution. (a): Suppose $(f_n)_{n=1}^\infty$ is a Cauchy Sequence of functions in $C^\alpha([0, 1])$. $\forall \varepsilon > 0$ there exists a $N \in \mathbb{N}$ such that $m \geq n \geq N$ we have $\|f_n - f_m\|_{C^\alpha} < \varepsilon$. $\forall x \in [0, 1]$,

$$|f_n(x) - f_m(x)| < \|f_n - f_m\|_{C^\alpha} < \varepsilon$$

Thus the sequence $(f_n(x))_{n=1}^\infty$ is Cauchy in $\mathbb{R}$. By completeness of $\mathbb{R}$, the limit exists. Let $f : [0, 1] \rightarrow \mathbb{R}$ where

$$f(x) = \lim_{n \rightarrow \infty} f_n(x).$$

1. $f_n \rightarrow f$ with respect to $\|\cdot\|_{C^\alpha}$: $\forall \varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for $m \geq n \geq N$ we have that $\|f_n - f_m\|_{C^\alpha} < \frac{\varepsilon}{4} \Rightarrow \|f_n - f_m\|_\infty < \frac{\varepsilon}{4}$. Thus $\forall x \in [0, 1]$,

$$|f_n(x) - f(x)| = \lim_{m \rightarrow \infty} |f_n(x) - f_m(x)| \leq \frac{\varepsilon}{4}$$

Thus $\|f_n - f\|_\infty < \frac{\varepsilon}{4}$. $\|f_n - f_m\|_{C^\alpha} < \frac{\varepsilon}{4}$ also gives us that for $x, y \in [0, 1]$ where $x \neq y$

$$\frac{|f_n(x) - f_m(x) - f_n(y) + f_m(y)|}{|x - y|^\alpha} \leq \frac{\varepsilon}{4}.$$

Thus do the same thing as above and we can bound the secants of $f_n - f$. Thus $\forall n \geq N$ we have that

$$\|f_n - f\|_{C^\alpha} \leq \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2} < \varepsilon.$$

2. $f \in C^\alpha([0, 1])$: Since $f_n \rightarrow f$ with respect to the $\|\cdot\|_{C^\alpha}$, there exists a N such that $\|f_N - f\|_{C^\alpha} < 1$.

$$\|f\|_{C^\alpha} \leq \|f_N\|_{C^\alpha} + \|f_N - f\|_{C^\alpha} < \|f_N\|_{C^\alpha} + 1 < \infty.$$

(b): We want to first show that $f_n \in C^\alpha([0, 1])$ for all $n \in \mathbb{N}$ for $\alpha \leq \frac{1}{2}$. To do this we need to show that $x, y \in [0, 1]$ where $x \neq y$

$$\frac{|f(x) - f(y)|}{|x - y|^{\frac{1}{2}}} < 1 \Rightarrow \frac{|f(x) - f(y)|}{|x - y|^\alpha} < 1$$

Note the following for $x, y \in [0, 1]$, $0 \leq |x - y| \leq 1$. Thus $|x - y|^{p_1} > |x - y|^{p_2}$ if $p_1 < p_2$. Thus

$$\frac{|f(x) - f(y)|}{|x - y|^\alpha} < \frac{|f(x) - f(y)|}{|x - y|^{\frac{1}{2}}} \leq 1.$$

Thus $f_n \in C^\alpha([0, 1])$.

Note the following about the sequence. $\forall n \in \mathbb{N}$, $\forall x, y \in [0, 1]$ where $x \neq y$ we have that

$$\frac{|f_n(x) - f_n(y)|}{|x - y|^{\frac{1}{2}}} \leq 1 \Rightarrow |f_n(x) - f_n(y)| \leq |x - y|^{\frac{1}{2}}.$$

Thus for $\varepsilon > 0$ and for all $n \in \mathbb{N}$, we have that if $\delta = \varepsilon^2$ then for $|x - y| < \delta$ we have that

$$|f_n(x) - f_n(y)| \leq |x - y|^{\frac{1}{2}} < (\varepsilon^2)^{\frac{1}{2}} = \varepsilon.$$

Thus $\{f_n\}_{n=1}^\infty$ is a equicontinuous sequence. f_n also bounded above by 1. Thus by Arzela Ascoli, there exists a subsequence which converges uniformly in $C([0, 1])$. $f_n \rightarrow f$ in $C([0, 1])$ means its uniformly cauchy. Thus $\forall \varepsilon$ there exists a $N \in \mathbb{N}$ such that for all $m \geq n \geq N$ we have that

$$2\|f_n - f_m\|^{1-2\alpha} < \frac{\varepsilon}{4}$$

$$\frac{|(f_m - f_n)(x) - (f_m - f_n)(y)|}{|x - y|^\alpha} = |(f_m - f_n)(x) - (f_m - f_n)(y)|^{1-2\alpha} \left(\frac{|(f_m - f_n)(x) - (f_m - f_n)(y)|}{|y - x|^{\frac{1}{2}}} \right)^{2\alpha}$$

We have that

$$\begin{aligned} |(f_m - f_n)(x) - (f_m - f_n)(y)|^{1-2\alpha} &\leq (|f_m(x) - f_n(x)| + |(f_m - f_n)(y)|)^{1-2\alpha} \\ &= (2\|f_m - f_n\|_\infty)^{1-2\alpha} \end{aligned}$$

We also have

$$\frac{|(f_m - f_n)(x) - (f_m - f_n)(y)|}{|x - y|^{\frac{1}{2}}} \leq \frac{|f_m(x) - f_m(y)| + |f_n(x) - f_n(y)|}{|x - y|^{\frac{1}{2}}} \leq 2$$

Thus

$$\frac{|(f_m - f_n)(x) - (f_m - f_n)(y)|}{|x - y|^\alpha} \leq (2\|f_m - f_n\|_\infty)^{1-2\alpha} \times 2^{2\alpha} = 2\|f_m - f_n\|_\infty^{1-2\alpha} < \frac{\varepsilon}{4}.$$

Thus $\|f_m - f_n\|_{C^\alpha} \leq \frac{\varepsilon}{2} < \varepsilon$. Thus by completeness of $C^\alpha([0, 1])$ and uniqueness of limit $f_n \rightarrow f$ in $C^\alpha([0, 1])$.

Problem 2 : Polynomial Approximation on Unbounded Sets

Let $E \subseteq \mathbb{R}$ be unbounded and let $f \in C(E)$ be such that for every $\varepsilon > 0$ there exists a polynomial p such that

$$|f(x) - p(x)| \leq \varepsilon, \quad x \in E.$$

Show that f is a polynomial.

Solution. Let $p_n \in C(E)$ be a sequence of polynomials such that $\|f - p_n\|_\infty < \frac{1}{n}$. Uniform convergence implies uniform Cauchy. $\forall \varepsilon > 0$, there exists a $N \in \mathbb{N}$ such that $m \geq n \geq N$ where $\|p_n - p_m\|_\infty < \varepsilon$. Let $Z := \{x \in E : p_N(x) = 0\}$.

$$|p_m(z) - p_N(z)| = |p_m(z)| < \varepsilon$$

Problem 3 : Interpolating Polynomial Approximation

Let $x_1, \dots, x_n$ be a finite collection of distinct points in $[0, 1]$ and let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous. Show that there exists a sequence of polynomials agreeing with f at each point x_k , $1 \leq k \leq n$, which converges to f uniformly on $[0, 1]$.

Hint. Introduce the Lagrange interpolating polynomials

$$L_k(x) = \prod_{j \neq k} \frac{x - x_j}{x_k - x_j}, \quad 1 \leq k \leq n.$$

Solution.

Problem 4 : Approximation on the Quarter Disc

Let $H = \{(x, y) \in \mathbb{R}^2; x, y \geq 0, x^2 + y^2 \leq 1\}$.

(a) Prove that for each $\varepsilon > 0$ and each continuous function $f : H \rightarrow \mathbb{R}$ there exists a function g of the form

$$g(x, y) = \sum_{m, n=0}^N a_{mn} x^{2m} y^{2n}, \quad N \in \mathbb{N},$$

$$a_{mn} \in \mathbb{R},$$

such that $|f(x, y) - g(x, y)| \leq \varepsilon$ for all $(x, y) \in H$.

(b) Does the result in part (a) hold if H is replaced by the disc $D = \{(x, y) \in \mathbb{R}^2; x^2 + y^2 \leq 1\}$?

Solution.

Problem 5 : Uniform Boundedness via Quadratic Inequality

Let $f : [0, \infty) \rightarrow [0, \infty)$ be continuous with $f(0) = 0$. Show that if

$$f(t) \leq 1 + \frac{1}{10} f(t)^2, \quad t \geq 0,$$

then f is uniformly bounded throughout $[0, \infty)$.

Solution.

Let $E = \{x \in [0, \infty) : f(x) \leq M\}$, $0 \in E$. We want to show that E is both open and closed. $f^{-1}([0, 2]) = E$ is closed. We want to show that E is open. Fix $x \in E$.

By continuity of f there exists a $\delta > 0$ such that $y \in (x - \delta, x + \delta)$:

$$|f(y) - f(x)| < \varepsilon \Rightarrow f(y) \leq \varepsilon + f(x) \leq \varepsilon + M$$

$$f(y) \leq 1 + \frac{1}{10}(\varepsilon + M)^2 = 1 + \frac{1}{10}(M^2 + 2M\varepsilon + M)$$